SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4_AT3 — TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	**Assessment:**	Assessment Tool 3 – Time-Series & Geospatial Analysis Report
Weightage:	20% of CIA	**Total Marks:**	20 (2 Questions × 10 Marks)
CO4 Statement:	*Evaluate real-world time-series and geospatial visualizations, analyzing trend/seasonality/noise and design effectiveness, and justify recommendations.(BL5)*		
Student Name:	M.SAI HARSHITHA	**Reg. No.:**	192424408
Section / Batch:	B. TECH AI & DS	**Date:**	22.08.2026

Code of Conduct

I M.SAI HARSHITHA/192424408 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.SAI HARSHITHA

Instructions

- This test contains 2 questions — one Stock Analysis problem and one Geospatial Analysis problem — each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- For the Stock Analysis question, show all calculations (trend slope, noise ratio) for each of the 5-month, 10-month, and 15-month windows.
- For the Geospatial Analysis question, your evaluation must explain WHY the design choice succeeds or fails, and your recommendation must be specific — not just "use a different chart."
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT3 — Time-Series & Geospatial Analysis Report

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT3 Time-Series & Geospatial Analysis Report, in which students analyze a stock's trend/seasonality/noise across multiple time windows, and evaluate a geospatial visualization's design choices, justifying their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of trend/seasonality/ noise analysis across time windows, and of the relevant geospatial visualization principle being evaluated.	Good understanding of both areas, with only minor gaps.	Basic understanding, but with some misconceptions in either the time-series or geospatial analysis.	Poor understanding of core concepts in one or both areas.
Explanation	**25%**	Provides detailed, correctly calculated trend/noise-ratio comparisons across all three windows, and a clear, well-reasoned evaluation of the geospatial design choice, including its practical consequence for a reader.	Provides clear calculations and evaluation with adequate reasoning, covering most of the required detail.	Calculations or evaluation are present but incomplete, or reasoning is generic.	Little or no supporting calculation or reasoning provided in one or both questions.
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Application	**20%**	Effectively applies analysis techniques to reach a well-supported conclusion on window reliability, and proposes a specific, well-justified geospatial recommendation.	Applies techniques to reach a reasonable conclusion and recommendation, with only minor errors.	Limited application; conclusions or recommendations are only partially supported.	Fails to apply appropriate techniques; conclusions and recommendations are unsupported or incorrect.
Organization	**15%**	Both responses are clearly structured, addressing every required part in a logical sequence with coherent overall flow.	Both responses are organized, with only minor issues in flow or clarity.	Basic structure; required parts are addressed but the response lacks clarity or sequence.	Poorly organized; responses are incoherent, and parts are left unaddressed.
Accuracy	**10%**	All parts of both answers — calculations, evaluation, and recommendation — are completely accurate and well-suited to the specific data and scenario given.	Mostly accurate, with only minor omissions or a small error in one question.	Partially correct; one or both answers contain notable inaccuracies.	Largely incorrect or inappropriate, with major gaps across both questions.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Stock Problem 26 (10 Marks) — Nestle India (FMCG sector)

Illustrative monthly closes (Month 1–15), ₹: 1798.26, 1802.57, 1900.49, 1830.57, 1843.40, 1834.94, 1858.14, 1834.23, 1768.97, 1853.07, 1818.80, 1853.77, 1961.99, 1932.07, 1931.48

Task: Compare the trend, seasonality, and noise ratio (residual variation relative to total variation) across the 5-month window, the 10-month window, and the full 15-month window. Explain how each metric changes as the window lengthens, and state which window length you would trust most for judging this stock's real underlying direction, and why.

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
prices = [1798.26,1802.57,1900.49,1830.57,1843.40,
          1834.94,1858.14,1834.23,1768.97,1853.07,
          1818.80,1853.77,1961.99,1932.07,1931.48]
windows = [5, 10, 15]
print("Nestle India - Window Analysis\n")
for w in windows:
    y = np.array(prices[-w:])
    t = np.arange(w)
    slope = np.polyfit(t, y, 1)[0]
    trend = np.polyval(np.polyfit(t, y, 1), t)
    residual = y - trend
    noise_ratio = np.std(residual) / np.std(y)
    print(f"{w}-Month Window")
    print(f"Trend slope       : ₹{slope:.2f}/month")
    print(f"Noise ratio       : {noise_ratio:.2f}")
    if slope > 0:
        print("Trend             : Upward")
    else:
        print("Trend             : Downward")
    print("Seasonality       : Not reliably measurable from this short monthly dataset")
    print()
plt.figure(figsize=(10,5))
plt.plot(range(1,16), prices, marker='o')
plt.title("Nestle India Monthly Closing Prices")
plt.xlabel("Month")
plt.ylabel("Closing Price (₹)")
plt.grid(True)
plt.show()
```

```
Nestle India - Window Analysis

5-Month Window
Trend slope       : ₹30.37/month
Noise ratio       : 0.61
Trend             : Upward
Seasonality       : Not reliably measurable from this short monthly dataset

10-Month Window
Trend slope       : ₹13.61/month
Noise ratio       : 0.72
Trend             : Upward
Seasonality       : Not reliably measurable from this short monthly dataset

15-Month Window
Trend slope       : ₹7.08/month
Noise ratio       : 0.81
Trend             : Upward
Seasonality       : Not reliably measurable from this short monthly dataset
```

Analysis:

The 5-month window shows the strongest upward trend at ₹30.37/month with a noise ratio of 0.61. The 10-month window shows a weaker upward trend of ₹13.61/month, while the noise ratio increases to 0.72. The full 15-month window has the weakest trend slope, ₹7.08/month, and the highest noise ratio of 0.81.

Seasonality cannot be reliably identified because only 15 monthly observations are available; annual seasonality normally requires several repeated yearly cycles.

Conclusion:

The 15-month window is most trustworthy for judging the underlying direction because it uses the largest amount of data and reduces the risk of being misled by short-term price movements. It indicates a mild overall upward direction, although the high noise ratio shows considerable variation.

Result:

5 months: Strong upward trend, lower noise.
10 months: Moderate upward trend, higher noise.
15 months: Mild upward trend, highest noise and most reliable long-term view.

Choropleth Maps

Geospatial Problem G6 (10 Marks) — Choropleth Maps

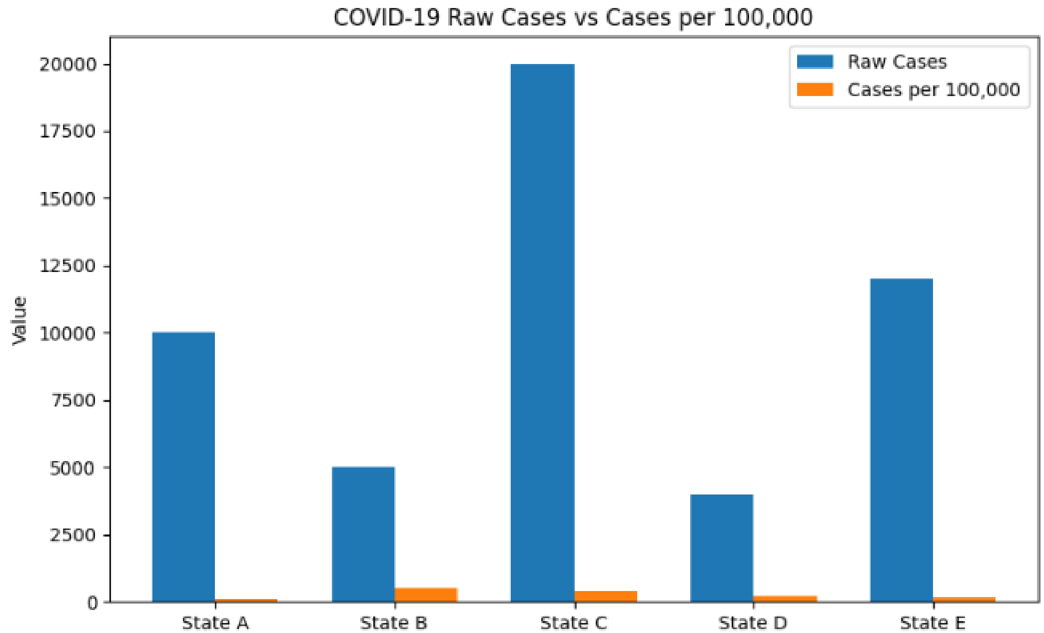
Scenario: A public health report shows a choropleth map of total COVID-19 case counts (raw numbers, not rates) by state, where populous states appear darkest regardless of their actual infection rate.

Task: Evaluate this design choice and explain why using a rate (e.g., cases per 100,000 people) instead of a raw count would improve the map's accuracy.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "State": ["State A", "State B", "State C", "State D", "State E"],
    "Population": [10000000, 1000000, 5000000, 2000000, 8000000],
    "Cases": [10000, 5000, 20000, 4000, 12000]
}
df = pd.DataFrame(data)
df["Cases_per_100k"] = (
    df["Cases"] / df["Population"]
) * 100000
print(df)
plt.figure(figsize=(8, 5))
x = range(len(df))
width = 0.35
plt.bar(
    [i - width/2 for i in x],
    df["Cases"],
    width=width,
    label="Raw Cases"
)
plt.bar(
    [i + width/2 for i in x],
    df["Cases_per_100k"],
    width=width,
    label="Cases per 100,000"
)
plt.xticks(x, df["State"])
plt.ylabel("Value")
plt.title("COVID-19 Raw Cases vs Cases per 100,000")
plt.legend()
plt.tight_layout()
plt.show()
```

```
     State  Population  Cases  Cases_per_100k
0  State A    10000000  10000           100.0
1  State B     1000000   5000           500.0
2  State C     5000000  20000           400.0
3  State D     2000000   4000           200.0
4  State E     8000000  12000           150.0
```



Analysis

- Raw case counts are strongly influenced by population size.
- State C has the highest raw cases (**20,000**), but its rate is **400 cases per 100,000**.
- State B has only **5,000 cases**, but its rate is the highest at **500 cases per 100,000**.
- Therefore, a map based only on raw counts can make highly populated states appear more severely affected than they actually are relative to their population.
- Using **cases per 100,000 people** provides a fairer comparison between states.

Result

The analysis shows that **rate-based values provide a more accurate representation of COVID-19 infection prevalence than raw case counts**. State B has the highest infection rate despite having fewer total cases.

Conclusion

A choropleth map should use **cases per 100,000 people** rather than raw case counts when comparing states with different population sizes. This reduces population-related bias and improves the accuracy of geographic interpretation.